Appl. Numer. Math. 197 (2024) 186–209

Contents lists available at ScienceDirect

Applied Numerical Mathematics

journal homepage: www.elsevier.com/locate/apnum

ELSEVIER

Error estimates for the finite element method of the Navier-Stokes-Poisson-Nernst-Planck equations ⋆

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ARTICLE INFO

Keywords:
NSPNP equations
Unconditional stability
Unconditional error estimates

ABSTRACT

In this paper, we consider the linearized backward Euler finite element scheme for the Navier-Stokes-Poisson-Nernst-Planck (NSPNP) equations. We aim to derive the unconditional error estimates of all variables in $l^\infty(L^2)$ and $l^\infty(H^1)$ norms. Compared with some previous results, we remove the mesh restriction. In addition, we derive two important physical properties for the proposed numerical scheme. Finally, three numerical examples are implemented to verify the theoretical analysis.

1. Introduction

2. Preliminary

3. The linearized backward Euler scheme

4. The error estimates

4.1. The $l^\infty(L^2)$-norm error estimates

4.2. The $l^\infty(H^1)$-norm error estimate

4.3. Improved error estimate for the $l^\infty(L^2)$-norm

5. Two physical properties

6. Numerical implementation

7. Conclusion and future work

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⋆ This work was supported by Cultivation Programme for Young Backbone Teachers in Henan University of Technology, Natural Science Foundation of Henan Province (222300420143, 232300420117, 232300420118), and the Innovative Funds Plan of Henan University of Technology (2021ZKCJ11).

https://doi.org/10.1016/j.apnum.2023.11.012

Received 14 February 2023; Received in revised form 27 August 2023; Accepted 16 November 2023

Available online 20 November 2023

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